



Spatiotemporal trends and drivers of evapotranspiration across india's diverse climatic zones

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Abstract

Evapotranspiration (ET) is a crucial component of the water cycle, significantly impacting climate regulation, hydrological processes, agriculture, and water resource management. This review offers a comprehensive analysis of ET trends across various climatic zones in India, examining regional variations, measurement methodologies, and underlying causes. The study highlights seasonal and spatial fluctuations in ET, with some regions experiencing declining ET due to reduced solar radiation and land-use changes, while others show increasing trends driven by temperature rise and agricultural expansion. Factors such as monsoon variability, deforestation, irrigation practices, and urbanization have a significant influence on ET patterns across the country. Different measurement techniques, including the Penman-Monteith equation, remote sensing, and statistical trend analysis, are examined for their effectiveness in estimating ET. The study also discusses the implications of ET changes for water resource management, agricultural productivity, ecosystem health, and climate modelling. Given the complex interactions between ET, climate change, and land-use dynamics, the review underscores the need for advanced monitoring techniques, AI-driven modelling approaches, and region-specific adaptation strategies to ensure sustainable water resource management. Future research should focus on enhancing the accuracy of ET estimation, understanding the interactions between climate and ET, and developing predictive models for improved environmental and agricultural planning in India.

Keywords Evapotranspiration · Climatic zones · Water resource management · Land-use change · Remote sensing

1 Introduction

Evapotranspiration (ET) is a crucial component of the water cycle and agricultural water balance, encompassing the combined processes of water loss from the Earth's surface through soil and water evaporation, as well as plant transpiration (Wanniarachchi and Sarukkalgige 2022). ET plays a

vital role in connecting terrestrial water, carbon, and surface energy exchanges (Zhang et al. 2016c). However, measuring and predicting ET, particularly at large spatial scales, remains inherently challenging (Zhang et al. 2016c). It is influenced by a range of factors, including soil moisture, vegetation cover, and climatic conditions. Notably, the relationship between ET and soil moisture during the drying phase is complex, as natural soil heterogeneity can modify the relationship between regional ET and average soil moisture (Wetzel and Chang 1987). Overall, ET is a fundamental hydrological process that returns approximately 60–80% of precipitation to the atmosphere, serving as a source for future precipitation (Tateishi and Ahn 1996). A clear understanding and accurate estimation of ET are essential for effective agricultural water resource planning, management, and regulation, as well as for evaluating water availability in terrestrial ecosystems (Zhou and Zhou 2009; Wanniarachchi and Sarukkalgige 2022). Given its significance in the global water cycle and its influence on human and socio-economic

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systems, there is a pressing need for improved datasets and estimation techniques.

1.1 Importance of evapotranspiration in the hydrological cycle

ET is a vital component of the hydrological cycle, influencing both the land surface water balance and the energy balance (Zhao et al. 2013). It encompasses the combined loss of water to the atmosphere through evaporation from soil and wet surfaces, as well as transpiration from plants (Pham et al. 2016). As such, ET plays a critical role in simulating the hydrological impacts of climate change and serves as a key parameter in water resource management practices, including irrigation scheduling and assessments of the hydrological effects (Zhao et al. 2013; Pham et al. 2016). Recent studies have emphasized the role of physiological responses to elevated atmospheric CO₂ in altering ET and evaporative fraction, potentially exerting a stronger influence on long-term runoff than radiative or precipitation changes (Lemordant et al. 2018). This finding highlights the tight coupling between the carbon and water cycles over land and underscores the importance of vegetation in shaping future terrestrial hydrologic responses. ET also contributes significantly to atmospheric moisture through recycling, which is essential for regional precipitation. For example, studies have found that up to 80% of summertime precipitation in parts of North America originates from land surface ET (Harrington et al. 2023). Therefore, understanding and accurately quantifying the components of ET, namely transpiration, canopy evaporation, and ground evaporation, is crucial for improving drought prediction models, managing water resources, and evaluating the hydrological impacts of land cover changes (Dimitriadou and Nikolakopoulos 2021; Harrington et al. 2023).

1.2 Significance of studying ET trends in India

Studying ET trends in India is essential due to their direct implications for hydrological processes, agricultural sustainability, and water resource planning. Analysis of evaporation time-series data from various meteorological stations in India has revealed a declining trend in both pan evaporation and potential evapotranspiration (PET) in recent decades, contrary to the expected increase under global warming scenarios (Chattopadhyay and Hulme 1997). Further analysis of PET trends during the second half of the 20th century showed significant increases in maximum and minimum temperatures as well as PET, particularly across the interior peninsular regions of India (Sonali and Nagesh Kumar 2016). These changes are driven by regional land surface processes and their feedback with climate systems.

ET also plays a crucial role in India's monsoon dynamics through the mechanism of moisture recycling. Northeastern India, with dense vegetation and high ET rates, exhibits a recycling ratio of approximately 25%—the highest in the country. This high recycling contributes to delayed monsoon withdrawal in September over central and northeastern India (Pathak et al. 2014). Therefore, monitoring ET trends in India is indispensable for understanding regional climate dynamics, formulating agricultural policies, and managing increasingly strained water resources. The intricate interactions between ET, temperature, precipitation, and land surface processes underscore the need for ongoing research in this domain to inform climate adaptation and water governance strategies.

1.3 Objectives of the review

The review of ET trends across different climatic zones in India aims to provide a comprehensive understanding of ET dynamics in the context of climatic diversity, hydrological variability, and climate change. The primary objectives are as follows:

1. Examine the spatiotemporal variations of ET across India's diverse climatic regions, ranging from arid deserts to humid forests. It includes analysing trends in ET rates and spatial patterns to identify regional heterogeneity (Vinnarasi et al. 2017; Mo et al. 2018).
2. Investigate the relationships between ET and key climate variables such as precipitation, temperature, and leaf area index (LAI), to understand the primary drivers of ET variability in different zones. For instance, ET variability has been shown to follow precipitation trends on an annual scale, while on a monthly scale, it correlates more strongly with net radiation (Zhang et al. 2014a; Mo et al. 2018).
3. Assess the relationship between ET trends and drought conditions using indices such as the Standardized Precipitation Evapotranspiration Index (SPEI). It helps evaluate water deficits and surpluses across different climatic regions and their implications for drought vulnerability (Rehana and Monish 2020; Um et al. 2020).
4. Explore the impacts of climate change on ET rates, with particular focus on variations in maximum and minimum temperatures that significantly influence ET patterns. Such an analysis is crucial for identifying climate-induced shifts in hydrological responses across different climatic zones (Adnan et al. 2017a; Vinnarasi et al. 2017).
5. Evaluate the performance of ET estimation methods, such as the Penman-Monteith equation, across India's diverse climatic conditions. It supports efforts to

improve ET modeling accuracy in both data-rich and data-scarce regions (Srinivas et al. 2013; Adnan et al. 2017b).

6. Understand the linkage between ET trends, water use efficiency, and carbon uptake by vegetation in different zones. This aspect is critical for ecosystem management, particularly in the context of climate-resilient agriculture and forest sustainability (Zhang et al. 2014a; Mo et al. 2018).

By addressing these objectives, the review seeks to enhance the understanding of ET dynamics across India's varied climatic regimes. Such insights are crucial for informing water resource management, drought preparedness, agricultural planning, and adaptation strategies in response to changing climatic conditions.

Figure 1 presents a comprehensive framework outlining the six key objectives of the ET review in India. The first objective focuses on analysing the spatiotemporal variability of ET across India's diverse climatic zones, ranging from hot arid deserts to humid tropical forests (Vinnarasi et al. 2017; Mo et al. 2018). It involves examining ET trends over multiple time scales to capture regional and seasonal differences, which is crucial for water planning and agricultural decision-making. The second objective examines the relationships between ET and key climatic variables, including precipitation, temperature, net radiation, and vegetation characteristics such as the leaf area index (Zhang et al. 2014a; Mo et al. 2018). Understanding these interdependencies helps in identifying the dominant drivers of ET variability in different regions, especially under changing climate conditions.

The third and fourth objectives assess the role of ET in drought monitoring and the influence of climate change on ET trends. Drought assessment is enhanced through indices such as the SPEI, which incorporates both water availability and atmospheric demand (Rehana and Monish 2020; Um et al. 2020). Climate change impacts are studied by evaluating

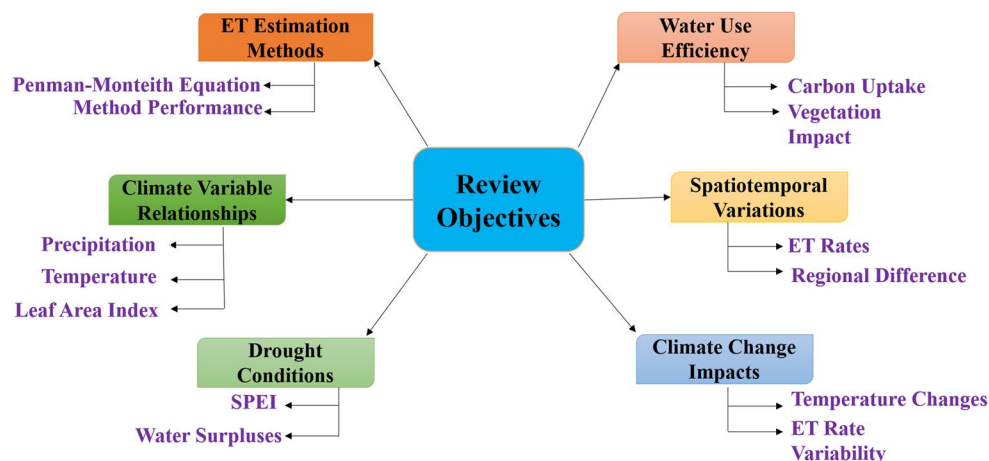
changes in temperature, CO₂ responses, and their combined effects on ET patterns and water stress (Adnan et al. 2017b; Vinnarasi et al. 2017). The fifth objective involves evaluating different methods used to estimate ET, including the Penman-Monteith equation and satellite-based approaches, to improve accuracy and applicability across various climatic zones (Srinivas et al. 2013; Adnan et al. 2017b). Finally, the sixth objective highlights the relationship between ET and ecosystem processes, such as water use efficiency and carbon uptake, emphasizing the importance of ET in sustainable resource management and climate-resilient development strategies (Zhang et al. 2014b; Mo et al. 2018).

2 Overview of India's climatic zones

2.1 Classification of climatic zones in India

India's diverse geography and topography result in a wide range of climatic zones across the country. Several studies have attempted to classify these zones using various methods and parameters. The bioclimatic classification of North-east India (NEI) identified three distinct zones: warm and humid, cool and humid, and cold and cloudy (Singh et al. 2007). This classification, based on over 30 years of meteorological data, is useful for climate-responsive building design and energy conservation. In contrast, a broader study of the entire Indian subcontinent revealed that the state of Karnataka alone encompasses five of the six major climate types found in India (Gunnell 1997). This diversity is attributed to the interaction between the Western Ghats Mountain barrier and the southwest monsoon, creating a steep environmental gradient where annual rainfall can drop from 6000 mm to 600 mm over just 80 km. Interestingly, different classification methods yield varying results. A study using machine learning algorithms identified 15 homogeneous rainfall regions in India, capturing the spatial variability of rainfall and its spell characteristics (Mannan et al. 2018).

Fig. 1 Objectives of the Evapotranspiration Review in India



This approach incorporated both rainfall data and climatic variables, revealing greater heterogeneity than classifications based solely on rainfall characteristics. Another study, which applied the Koppen system of climate classification to the state of Minas Gerais, Brazil, identified three major climatic groups (A, B, and C), corresponding to tropical rainy, dry, and warm-temperate climates (de Sá Júnior et al. 2012). While this study focused on Brazil, it demonstrates the applicability of the Koppen system to diverse regions. Therefore, the classification of climatic zones in India is complex and depends on the parameters and methods used. From broad categories such as warm-humid and cold-cloudy to more nuanced divisions based on rainfall patterns and other climatic variables, these classifications offer valuable insights for various applications, including agriculture, building design, and water resource management.

2.2 Characteristics of each climatic zone

India exhibits a range of climatic zones, each characterized by distinct features. Hot-dry zones experience high temperatures and low humidity, with cities like Ahmedabad representing this climate. These areas benefit significantly from passive cooling strategies, particularly direct evaporative cooling, which is highly effective in such conditions (Bhamare et al. 2020). Hot-humid regions, such as Chennai, are characterized by high temperatures and high humidity levels. These areas are prone to heat stress, with mortality risks increasing by 8.0% during “sweltering” days compared to “comfortable” days (Choudhary et al. 2023). Temperate zones offer more moderate conditions, while cold zones experience significantly lower temperatures. The composite climate, exemplified by Delhi, exhibits varied conditions throughout the year. Delhi showed a 5.9% increase in mortality risk during extreme heat events (Choudhary et al. 2023). Interestingly, NEI exhibits unique rainfall patterns compared to other regions. While Northwest India (NWI), Central India (CI), and South Peninsula India (SPIN) experience stronger rainfall during active monsoon years, NEI’s rainfall remains relatively constant regardless of monsoon strength (Zheng et al. 2016). In conclusion, India’s climatic diversity necessitates region-specific approaches to building design, agriculture, and public health. The varying impacts of heat stress and rainfall patterns across different zones highlight the importance of tailored strategies for each climatic region (Zheng et al. 2016; Choudhary et al. 2023).

2.3 Factors influencing regional climate patterns

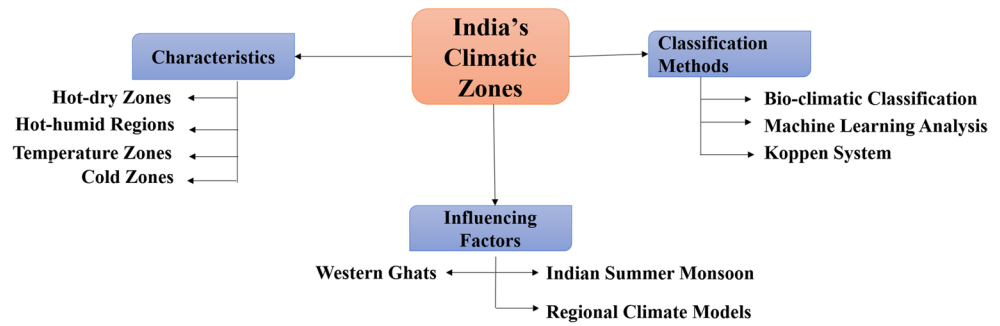
Regional climate patterns in India are influenced by a range of factors, with the Western Ghats Mountain barrier playing a particularly crucial role in shaping the climate of peninsular

India (Gunnell 1997). This 1,500-km-long mountain range acts as a climatic gatekeeper, creating a sharp environmental gradient on its leeward side, where annual rainfall drops dramatically from 6,000 mm to 600 mm over a distance of just 80 km. The Indian Summer Monsoon is a major driver of regional climate variability. Studies using regional climate models have shown that moisture availability over the Arabian Sea and the strength of low-level westerly winds are key contributors to precipitation changes in Southern India (Dessai et al. 2018). Additionally, the interaction between atmospheric structure and topographic relief contributes to the varied climate patterns across the country. For example, Karnataka alone contains five of India’s six major climate types due to its unique location and topographical diversity (Gunnell 1997). Interestingly, future climate projections show contradictory results. Some studies indicate an increase in the frequency of extreme wet events over CI under both near-future and far-future scenarios (Rai et al. 2019). Conversely, others predict significant deficits in precipitation in warmer climates, especially under higher emission scenarios (Kumar and Dimri 2018). These contrasting findings underscore the complexity and uncertainty inherent in regional climate projections. Hence, the interplay between topography, monsoon circulation, and large-scale climate drivers creates diverse and complex climate patterns across India. The Western Ghats have a significant influence on the climate of peninsular India, while the Indian Summer Monsoon remains the dominant factor affecting precipitation. Continued research and advanced modelling are crucial for a deeper understanding and more accurate prediction of future regional climate scenarios across the subcontinent.

Figure 2 illustrates the classification and influencing factors of India’s diverse climatic zones. India’s geography encompasses a diverse array of topographies, ranging from the Himalayas in the north to coastal plains and plateaus in the south, which contribute to the formation of multiple climatic zones, including hot arid, hot semi-arid, warm-humid, and cold climates (Gunnell 1997; Singh et al. 2007). Traditional classifications have been complemented by modern methods, such as machine learning and remote sensing, which enable more granular identification of homogeneous climatic regions based on parameters like rainfall, temperature, and vegetation (Mannan et al. 2018). For example, the northeastern region is characterized by warm and humid conditions, while northwestern India shows cold-arid or semi-arid traits. The Western Ghats play a pivotal role in modulating regional rainfall distribution by acting as a natural orographic barrier, which influences the steep rainfall gradient and forms microclimates within short spatial extents (Gunnell 1997).

Additionally, regional climate variability in India is heavily influenced by large-scale atmospheric systems such

Fig. 2 India's Climatic Zones: Classification and Influencing Factors



as the Indian Summer Monsoon and interactions with geographical features. The monsoon determines seasonal rainfall patterns, with studies indicating that parameters like low-level jet stream intensity and sea surface temperatures in the Arabian Sea significantly control monsoon-driven precipitation, particularly in peninsular India (Dessai et al. 2018). Anthropogenic influences and future climate change projections further amplify the complexity of India's climate. Some studies forecast an increase in extreme rainfall events in CI, while others predict a precipitation deficit in warmer regions under higher emission scenarios (Kumar and Dimri 2018; Rai et al. 2019). Thus, understanding the classification and drivers of India's climatic zones is crucial for adaptive planning in agriculture, water resource management, and infrastructure development, particularly in light of increasing climate uncertainties.

3 Methodology for ET measurement and trend analysis

3.1 Common methods for measuring evapotranspiration

ET measurement and trend analysis employ various methods, each with its strengths and limitations: The FAO-56 Penman-Monteith method is widely used for calculating reference ET (RET) and analysing trends. This approach was applied in Togo to explore RET trends from 1961 to 2011, revealing significant increases at some sites and a declining trend in the precipitation/RET ratio, indicating an increase in irrigation (Koffi and Komla 2015). The Penman-Monteith method is also used in combination with water balance approaches, as demonstrated in a study across China that analysed the temporal trends and spatial distribution of actual ET (AET) from 1960 to 2002 (Gao et al. 2007). Interestingly, while the Penman-Monteith method is considered reliable, it is parameter-rich and not always applicable in data-scarce regions. To address this, some researchers have developed simplified models based on multiple linear regression using readily available climatic data, such as solar radiation, air temperature, relative humidity, and

wind speed (Yirga 2019). Remote sensing provides another cost-effective method for estimating ET at regional to global scales, offering advantages in data-sparse areas (Zhang et al. 2016c). For more localized measurements, weighing lysimeters can provide precise daily ET estimates. At the same time, the precipitation-minus-stream discharge method can yield good annual ET totals, albeit with less precision on a monthly basis (Holmes 1984). Comparative studies have shown that satellite methods and FAO-24 methods tend to have the greatest variability, while FAO-56, hydrological models, and field methods show more consistency (Kite and Droogers 2000). In conclusion, the choice of ET measurement and trend analysis method depends on the availability of data, the scale of application, and the specific research objectives. While the Penman-Monteith method remains a standard, simplified approaches and remote sensing techniques are gaining importance, especially in data-limited regions.

3.2 Data sources and collection techniques

ET measurement and trend analysis rely on various data sources and collection techniques: Pan evaporation measurements have been commonly used, but are prone to errors that limit their reliability for trend analysis. A study in South Florida found inconsistencies in magnitude and direction between spatially related pan stations, concluding that pan evaporation data are insufficient for accurate trend analysis (Abtew et al. 2011). More robust approaches combine remote sensing data with ground-based meteorological measurements. Satellite and airborne sensors provide information on reflected solar radiation and surface temperatures, while ground stations measure other variables required for the analysis. It allows evaluation of ET at both local and regional scales (Jackson 1985). Weather station data is also used to calculate RET using methods such as the FAO-56 Penman-Monteith and adjusted Hargreaves equations (Gocic and Trajkovic 2014). Advanced techniques include using satellite soil moisture retrievals and meteorological data to partition ET into soil evaporation and transpiration components (Rigden et al. 2018). Lysimeters, eddy covariance systems, Bowen ratio methods, water balance approaches,

sap flow measurements, scintillometer, and satellite-based remote sensing are also employed; however, each has potential biases that require careful experimental design (Allen et al. 2011). In summary, while traditional methods, such as pan evaporation, have limitations, combining remote sensing, ground measurements, and advanced modelling techniques enables more comprehensive estimation of ET and trend analysis across various scales. However, researchers must be aware of potential biases in different measurement systems to ensure the quality and validity of their data.

3.3 Statistical approaches for trend analysis

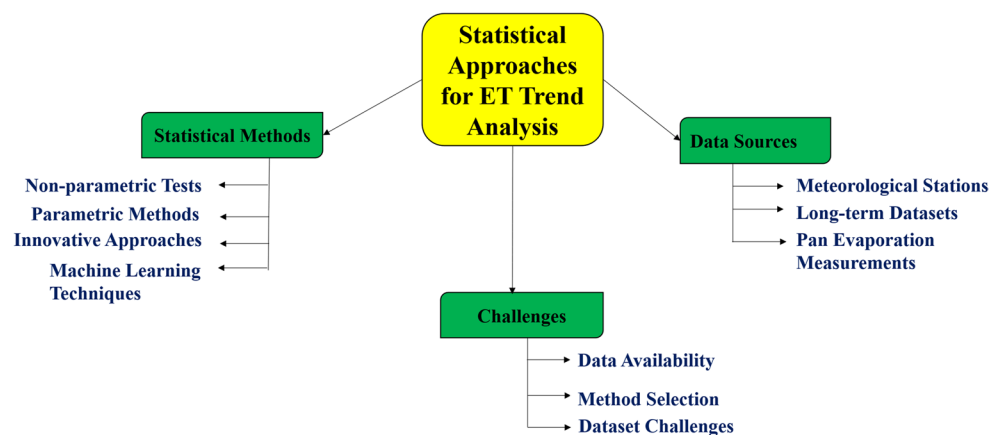
ET measurement and trend analysis rely on various data sources and statistical approaches. Data sources for ET analysis typically include meteorological stations that provide measurements of temperature, humidity, wind speed, and solar radiation (Abtew et al. 2011; Gocic and Trajkovic 2014). Some studies utilize long-term datasets spanning several decades to capture climate trends (Gocic and Trajkovic 2014; Pingale et al. 2016). Additionally, pan evaporation measurements have been used, though their reliability for trend analysis is questioned due to potential errors (Abtew et al. 2011). Statistical approaches for ET trend analysis commonly employ non-parametric tests such as the Mann-Kendall (MK) test, Modified Mann-Kendall (MMK) test, and Spearman's Rho test (Gocic and Trajkovic 2014; Pingale et al. 2016; Acar 2024). These methods are often applied at monthly, seasonal, and annual time scales to detect statistically significant trends. Some studies also employ parametric methods, such as linear regression (Gocic and Trajkovic 2014). Innovative approaches, such as Innovative Trend Analysis (ITA) and Innovative Polygon Trend Analysis (IPTA), have been introduced to complement traditional methods (Acar 2024; Gaddikeri et al. 2024). Interestingly, machine learning techniques such as Gene Expression Programming (GEP) and Random Forest (RF) have been applied to ET estimation and spatial interpolation, yielding promising results in some cases (da Silva Júnior et al. 2019).

However, the proper selection of training and testing datasets remains a challenge for these approaches. In conclusion, ET measurement and trend analysis involve a diverse range of data sources and statistical methods. While traditional non-parametric tests remain widely used, innovative and machine learning approaches are gaining traction. The choice of method depends on data availability, study objectives, and regional climate characteristics.

Figure 3 presents the statistical approaches commonly employed for analyzing ET trends, which are crucial for understanding hydrological changes and climate variability. One of the most widely used methods is the Mann-Kendall (MK) test, a non-parametric technique effective for detecting monotonic trends in time series data without requiring the data to conform to any specific distribution (Kendall 1948; Wang et al. 2004). The Sen's slope estimator is often used in conjunction with the MK test to quantify the magnitude of detected trends, providing a robust estimate of the rate of change in ET over time (Sen 1968). These tools have proven particularly useful in long-term climate studies across varied geographic and climatic zones, including in the Indian subcontinent. They enable researchers to examine trends over different temporal scales, such as annual, seasonal, and monthly intervals, thus offering nuanced insights into ET variability.

Additionally, correlation and regression analyses are used to assess the relationships between ET and climate variables, including temperature, precipitation, solar radiation, and relative humidity. Principal Component Analysis (PCA) and Empirical Orthogonal Functions (EOF) are multivariate techniques that help identify dominant spatial and temporal patterns in ET datasets, especially when working with large-scale gridded data from remote sensing or reanalysis products (Li et al. 2017b). The integration of machine learning algorithms, such as random forests and support vector machines, is increasingly being explored for ET modelling and prediction due to their ability to capture complex, nonlinear interactions between variables (Hosseini et al. 2020). These advanced approaches complement

Fig. 3 Statistical Approaches for Evapotranspiration Trend Analysis



traditional statistical tools, offering higher predictive accuracy and adaptability in dynamic climate systems. The combined use of these statistical methodologies enhances the ability to detect, explain, and model ET trends, which is essential for planning water resource management and mitigating climate-related risks.

4 ET trends in different climatic zones

4.1 Tropical wet (southwest monsoon) zone

ET trends in tropical wet (southwest monsoon) zones exhibit complex patterns influenced by various factors. In tropical monsoon environments, seasonal variations in ET are observed between the wet and dry seasons. During the monsoon season, latent heat flux dominates, corresponding to increased specific humidity after frequent precipitation. In contrast, sensible heat flux increases with surface temperature during the dry season (Toda et al. 2002). The daily integrated latent heat flux was found to be 5.18 MJ m^{-2} , accounting for a significant portion of the energy balance. Interestingly, some studies have found contrasting ET trends in different monsoon regions. In southwest China, where the region is influenced by the Pacific and Indian Ocean monsoons, ET generally increased in the Pacific monsoon region but decreased in the Indian Ocean monsoon region. This difference was attributed to varying climate conditions and vegetation responses (Sun et al. 2022). Similarly, in mainland Southeast Asia, seasonal patterns of ET differed among forest types, with some evergreen forests showing increased ET corresponding to atmospheric evaporative demand, while other evergreen and deciduous forests did not exhibit this pattern (Tanaka et al. 2008). In conclusion, ET trends in tropical wet zones are influenced by complex interactions between climate factors, vegetation characteristics, and monsoon patterns. While some regions show increasing ET trends, others experience decreases, highlighting the need for location-specific analyses. These trends have significant implications for water resource management, agriculture, and ecosystem functioning in monsoon-affected regions (Tanaka et al. 2008; Cui et al. 2020).

4.2 Tropical dry zone

Trends in ET in tropical dry zones exhibit complex patterns influenced by various climatic factors. Studies indicate that accounting for changes in ecosystem water demand is crucial when assessing dry season length and timing in the tropics (Xu et al. 2022). When considering AET, a larger fraction (48.7%) of tropical land is experiencing longer dry seasons, with Southern Amazonia and central Africa

identified as hotspots of dry season lengthening (Xu et al. 2022). Interestingly, the dynamics of ET are affected differently by various climatic variables across regions. While precipitation variability dominantly controls global ET in dry climates, Vapor Pressure Deficit (VPD) has a significant impact on ET trends in boreal mid-latitude climates (Xu et al. 2022). It highlights the importance of considering multiple factors when analysing ET trends in tropical dry zones. In conclusion, ET trends in tropical dry zones are complex and region-specific. The lengthening of dry seasons in some areas, coupled with the varying influences of precipitation, temperature, and VPD on ET, underscores the need for comprehensive approaches in studying and managing water resources in these regions. Future research should focus on integrating multiple climatic variables and their interactions to better understand and predict ET trends in tropical dry zones.

4.3 Montane zone

ET trends in montane zones show complex patterns influenced by various climatic factors and topography. In mountainous regions, ET exhibits strong spatial variations due to changes in elevation, vegetation, soils, and snow cover conditions (Gurtz et al. 1999). Studies have revealed both increasing and decreasing trends in ET across different montane areas. In the mountain plateau zone (MPZ) of China, a significant increase in minimum temperature and a decrease in sunshine hours contributed most to the increase in RET from 1956 to 2015 (Fan et al. 2016). However, in the mountains of southwest China, a positive relationship has been observed between ET change and station altitude (Thomas 2000). Interestingly, the high spatial variation in montane zones necessitates careful consideration when modelling hydrological processes. A complex hydrological model applied to the alpine/pre-alpine basin of the River Thur in Switzerland demonstrated that AET exhibits greater variability in lower and medium-altitude zones compared to upper elevation zones. This variability is associated with limitations in available moisture in soil and surface depression storages, as well as the evaporative demand of local vegetation (Gurtz et al. 1999). These findings highlight the importance of considering altitude-dependent factors when studying ET trends in montane regions.

4.4 Arid and semi-arid zones

ET trends in arid and semi-arid zones show significant variations and are influenced by multiple factors. Studies indicate an overall increasing trend in ET across many arid and semi-arid regions. For instance, in Pakistan, mean annual ET increased in extremely arid and arid zones, correlating

positively with rising temperatures and solar radiation (Adnan et al. 2017b). Similarly, in Iran, an average increasing trend of 4.42 mm/year was observed over four decades, with significant increases at 30% of the studied stations (Tabari et al. 2012). Interestingly, some studies reveal contradictory findings. In Xinjiang, China, ET decreased from 2001 to 2015, while other vegetation indicators, such as gross primary productivity and leaf area index, increased (Zhou et al. 2022). It highlights the complex interplay of factors affecting ET in arid and semi-arid regions. Climate variables, particularly temperature and precipitation, play crucial roles in ET trends. Wind speed, solar radiation, and vapor pressure also significantly influence ET (Tabari et al. 2012; Adnan et al. 2017a). In conclusion, although the general trend indicates an increasing ET in arid and semi-arid zones, regional variations exist. Human activities have a substantial impact, accounting for over 72% of changes in ecosystem indicators, including ET, in certain areas (Zhou et al. 2022). The complex interactions between climate factors and human activities underscore the need for region-specific studies and management strategies to address water resource challenges in arid and semi-arid zones.

Figure 4 illustrates the spatial and temporal trends of ET across India's major climatic zones—Hot Arid, Hot Semi-Arid, Cold Arid, Cold Semi-Arid, and Humid regions. These zones display distinctive ET behaviours influenced by temperature, precipitation, vegetation cover, and soil moisture availability. In the Hot Arid and Hot Semi-Arid zones, higher temperatures and limited rainfall contribute to increased ET demand. However, declining pan evaporation trends have been reported in parts of these regions, likely due to decreased wind speeds and increased atmospheric

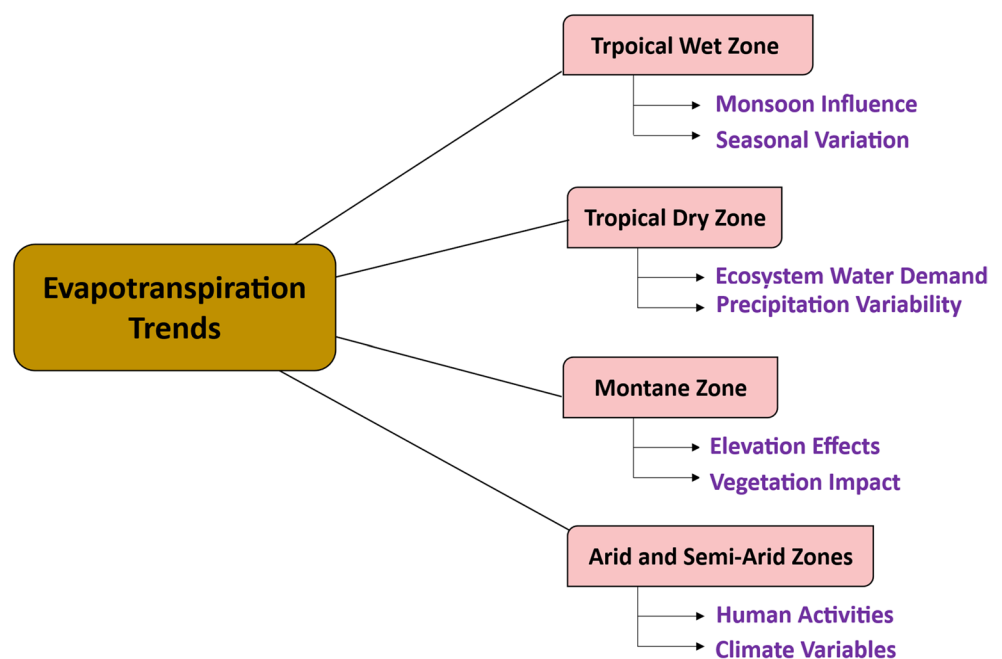
humidity (Chattopadhyay and Hulme 1997; Sonali and Nagesh Kumar 2016). The Cold Arid regions, particularly in the northwestern Himalayas, show low but gradually increasing ET trends due to rising temperatures and changes in snowmelt timing. Meanwhile, the Humid and Sub-Humid zones, such as northeastern and southern India, exhibit relatively stable ET trends, driven by dense vegetation and consistent rainfall patterns. Temporal analysis reveals that ET trends are not uniform across India and vary in response to climatic fluctuations and land-use changes. For instance, studies have shown an upward trend in PET in the central and interior peninsular regions of India, which correlates with increasing minimum and maximum temperatures (Adnan et al. 2017a; Vinnarasi et al. 2017). The leaf area index (LAI), a proxy for vegetation health, significantly influences ET variability, especially in monsoon-dependent agricultural zones (Zhang et al. 2014a; Mo et al. 2018). Moreover, the impacts of climate change are more pronounced in arid and semi-arid regions, where warming and soil moisture stress exacerbate ET losses, reducing water availability for crops. As such, continuous monitoring of ET trends is critical for assessing drought severity, improving irrigation practices, and supporting climate-resilient agricultural strategies across India's diverse climatic regions.

5 Regional variations in ET trends over India

5.1 North India

Regional variations in ET trends over North India exhibit complex patterns influenced by multiple factors. ET trends

Fig. 4 Evapotranspiration Trends in Climatic Zones



in North India show significant spatial and temporal variability. Studies indicate an overall increasing trend in ET and PET in recent decades, particularly in the Gangetic Plains region (Choudhury et al. 2021). This trend is more pronounced during the monsoon season, with significant increases observed in July and August (Choudhury et al. 2021). However, some studies have found decreasing trends in pan evaporation and PET in certain parts of India over the past few decades (Chattopadhyay and Hulme 1997). Interestingly, the relationship between AET and PET does not always follow expected patterns. The “Pan Evaporation Paradox” observed in some regions suggests that decreasing pan evaporation can occur alongside increasing AET, highlighting the complex interplay between different components of the water cycle (Hobbins et al. 2004). Additionally, the contribution of transpiration to total ET varies seasonally, with higher proportions during summer months and lower contributions in winter (Wei JianHui et al. 2015). Climate change is expected to significantly impact future ET patterns in North India. Projections indicate that increased temperatures will likely lead to higher PET in most parts of the country, although the magnitude of change will vary regionally (Chattopadhyay and Hulme 1997). Factors such as elevated CO₂ concentrations, land use changes, and aerosol deposition may also play important roles in modulating ET trends, potentially rivaling or exceeding the influence of climate in some areas (Shi et al. 2013). These complex interactions underscore the need for comprehensive approaches in water resource management and drought mitigation strategies for the region.

5.2 South India

Regional variations in ET trends have been observed in several studies over South India, with both increasing and decreasing trends reported across different areas and time periods. In northeast Louisiana, a study found that local crop coefficients (*K_c*) for cotton differed from FAO-adjusted values, highlighting the importance of regional variations in environmental conditions, cultivars, and management practices (Kumar et al. 2015). It highlights the need for locally developed tools to estimate crop water use and schedule irrigation. Interestingly, contradictory trends have been observed in different parts of India. While some regions in South India have experienced a decreasing trend in PET, others have shown an increase. A study analysing PET trends across China found that south of 35°N, sunshine was most strongly associated with changes in ET (Thomas 2000). In India, significant upward trends in maximum temperature, minimum temperature, and PET were observed over most grid points in the interior peninsular region

(Sonali and Nagesh Kumar 2016). In conclusion, the variability in ET trends over South India highlights the complex interplay of factors influencing these patterns. Factors such as changes in wind speed, relative humidity, temperature, and sunshine duration contribute to these regional variations (Chattopadhyay and Hulme 1997; Jhajharia et al. 2015). The use of AET in drought characterization has been proposed as a more robust approach for water-limited regions, such as parts of South India (Rehana and Monish 2020). These findings underscore the importance of considering regional factors when studying ET trends and their implications for water resource management and agricultural planning in South India.

5.3 East India

ET trends in East India show significant regional variations, influenced by seasonal patterns and climatic factors. In north-east India, a study using eddy covariance flux observations found that ET rates vary considerably across seasons, with maximum latent heat flux during the monsoon (300 W m⁻²) being double that of winter (150 W m⁻²) (Deb Burman et al. 2019). The Bowen ratio, which indicates the relative importance of sensible heat to latent heat, ranges from 0.93 in winter to 0.27 during the monsoon, highlighting the seasonal variability in ET processes. Interestingly, while some parts of eastern India show increasing trends in drought conditions due to decreasing rainfall, other areas exhibit different patterns. A study analysing the SPEI found significant increasing trends in both drought duration and magnitude over parts of eastern and northeastern India during the monsoon season (Das et al. 2016). This trend was primarily attributed to decreasing rainfall rather than changes in PET. In conclusion, ET trends in East India are characterized by strong seasonal variations and regional differences. The monsoon plays a crucial role in determining ET rates, with the highest values observed during this season. However, long-term trends indicate increasing drought conditions in some eastern regions, primarily driven by rainfall patterns rather than changes in PET. These findings underscore the complex interplay between precipitation, ET, and drought dynamics in the region, highlighting the need for comprehensive water management strategies that account for these spatial and temporal variations.

5.4 West India

Regional variations in ET trends over West India exhibit some interesting patterns. Decreasing trends in RET have been observed in NEI over a 22-year period, attributed mainly to decreases in net radiation and wind speed (Jhajharia et al.

2012). This trend counteracted the effects of temperature increases on RET in that humid region. In contrast, parts of western India, particularly the arid western region, exhibited significantly decreasing trends in meteorological drought conditions during the monsoon period from 1901 to 2008 (Das et al. 2016). It was primarily due to decreases in PET rather than increases in rainfall. Interestingly, ET from vegetation in the Western Ghats Mountain range, along India's western coast, was found to contribute significantly to monsoon rainfall in the water-deficient state of Tamil Nadu, further east (Paul et al. 2018). This contribution reached up to 50% during deficit monsoon years, highlighting the importance of forest cover in regional water cycles. In summary, while NEI has seen a decrease in ET trends, parts of western India have experienced reduced drought conditions due to lower evaporative demand. The Western Ghats forests play a crucial role in supplying moisture for rainfall in neighbouring regions, underscoring the importance of considering vegetation-climate feedbacks when assessing regional water balances and ET trends in India.

5.5 Central India

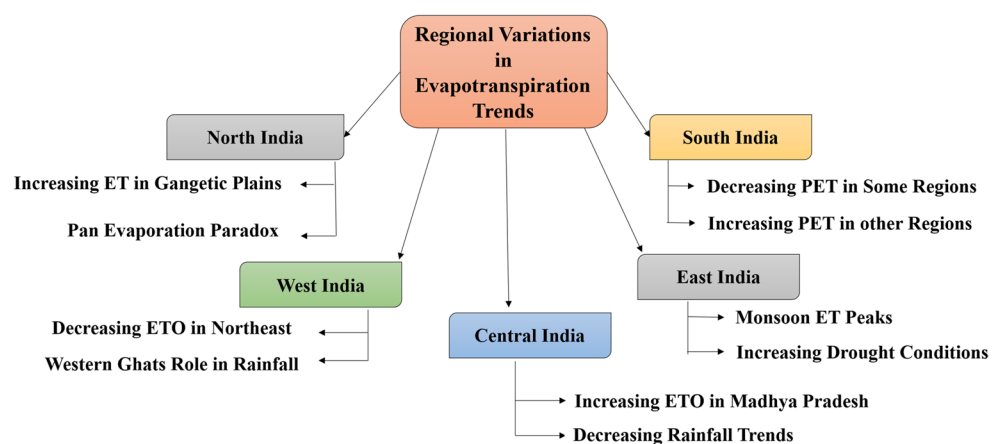
Regional variations in ET trends have been observed in several studies over CI. In northeast India, a significant decreasing trend in RET was observed at both annual and seasonal time scales over a 22-year period. This decrease was mainly attributed to declining net radiation and wind speed, which outweighed the impact of temperature increases (Jhajharia et al. 2012). Contrastingly, in the states of Madhya Pradesh and Chhattisgarh in CI, an increasing trend in RET was projected for the 21st century based on downscaled climate model data. The highest increases were projected for April, May, and October (Kundu et al. 2017). It aligns with findings of significant warming trends in Madhya Pradesh from 1951 to 21, particularly since the 1990s (Kumar et al. 2023). Interestingly, a study analysing rainfall trends in CI from 1901 to 2010 found that the majority of stations

examined in Madhya Pradesh and Chhattisgarh experienced mostly decreasing trends in annual, summer, and monsoon rainfall. However, the winter and post-monsoon seasons generally showed increasing trends (Sanikhani et al. 2018). In summary, ET trends appear to vary regionally within CI, with some areas showing decreases and others showing increases. These variations are likely driven by complex interactions between changing rainfall patterns, temperatures, radiation, and wind speeds. The contrasting trends highlight the importance of local and regional-scale analyses in understanding the impacts of climate change on ET in this region.

Figure 5 highlights the spatial heterogeneity in ET trends across different regions of India, emphasizing the influence of climatic, geographic, and land-use factors on these trends. The northwestern arid and semi-arid regions, particularly Rajasthan and Gujarat, exhibit complex ET trends. While PET is increasing due to rising temperatures, AET is often constrained by low soil moisture and limited vegetation cover (Chattopadhyay and Hulme 1997; Adnan et al. 2017a). In contrast, northeastern India displays relatively high and stable ET levels due to abundant rainfall and dense forest cover. This region also exhibits a strong vegetation-ET relationship, where higher leaf area index (LAI) significantly contributes to atmospheric moisture recycling (Pathak et al. 2014; Zhang et al. 2014a). The CI region, including parts of Madhya Pradesh and Chhattisgarh, shows upward ET trends, likely driven by increased cropping intensity and temperature rise, which enhances both transpiration and evaporation.

Meanwhile, southern peninsular India, particularly the interior parts of Karnataka, Andhra Pradesh, and Tamil Nadu, reveals mixed ET trends, some areas experiencing increases due to intensified irrigation and land cover changes, while others show reductions likely linked to declining groundwater levels and changing rainfall patterns (Vinnarasi et al. 2017; Bharghavi et al. 2025). Coastal zones with moderate humidity levels show relatively stable

Fig. 5 Regional Variations in Evapotranspiration Trends in India



ET patterns, benefiting from monsoonal precipitation and maritime influence. The Himalayan foothills and cold arid regions of Jammu & Kashmir and Ladakh report marginal increases in ET, primarily attributed to glacial melt and rising temperatures (Kumar and Dimri 2018). These findings highlight that ET responses are not uniform across India; instead, they are intricately tied to regional hydroclimatic systems. Understanding such regional variations is crucial for effective climate adaptation strategies, informed agricultural planning, and sustainable water resource management.

6 Underlying causes of ET trends

6.1 Climate change and global warming

ET trends are significantly influenced by climate change and global warming, with various factors contributing to these changes. Precipitation variability dominantly controls global ET in dry climates, while net radiation has substantial control over ET in tropical regions (Li et al. 2022). The vapor pressure deficit (VPD) influences ET trends in boreal mid-latitude climates, playing a crucial role in regulating the carbon-water-energy cycle (Li et al. 2022). Climate change has led to warming and drying trends, which in turn affect ET processes. For instance, in the Three Gorges Reservoir region, a decrease in relative humidity and an increase in wind speeds led to increased vegetation transpiration and ground evaporation, respectively (Wang et al. 2019). Interestingly, land use/land cover changes also drive ET changes in water bodies, urban areas, and high-altitude regions (Wang et al. 2019). The sensitivity of ET to global warming varies across different meteorological parameters. A 20% increase in temperature could lead to a 14.8% increase in total ET demand, while ET is less sensitive to increases in net solar radiation (11%) and wind speed (7%). Vegetation changes play a crucial role in ET trends, with increases in vegetation leaf area index, primarily driven by greening, leading to contrasting trends in transpiration and soil evaporation (Zhang et al. 2016b). These opposing trends underscore the importance of accurately representing vegetation changes in Earth system models for predicting future shifts in the energy and water cycles (Zhang et al. 2016b). In conclusion, the underlying causes of ET trends in the context of climate change and global warming are complex and interconnected. Factors such as precipitation variability, net radiation, VPD, temperature changes, and vegetation dynamics all contribute to the observed trends. Understanding these relationships is crucial for predicting future changes in the hydrological cycle and developing effective adaptation strategies (Kundzewicz 2008).

6.2 Land use and land cover changes

Land use and land cover changes (LULCC) have significant impacts on ET trends, affecting regional water cycles and ecosystem services. Several studies have identified key underlying causes of ET trends related to LULCC, with climate factors playing a crucial role in these trends. Temperature, precipitation, relative humidity, wind speed, and soil moisture are important drivers of ET variability (Su et al. 2023). In some cases, climate warming and increased rainfall variability have led to intensifying aridity conditions, affecting ET patterns (Suliman and Elagib 2012). The aridity index, representing the ratio of rainfall to RET, has shown values close to 1 in dry months, indicating medium to high aridity in some regions (de Oliveira et al. 2021). Vegetation changes due to LULCC significantly impact ET. Conversion of natural vegetation to agricultural land can alter ET rates (Suliman and Elagib 2012; Sulamo et al. 2021). Notably, NDVI has been identified as one of the dominant factors influencing ET changes in some areas (Su et al. 2023). Afforestation and woody encroachment tend to decrease albedo and increase ET, while urbanization and cultivation often increase albedo (Münch et al. 2019). Topography and soil characteristics also influence ET trends. Elevation is a significant factor affecting ET in some regions (Su et al. 2023). Additionally, changes in soil moisture due to LULCC can impact ET rates (Sulamo et al. 2021; Su et al. 2023). In conclusion, the underlying causes of ET trends concerning LULCC are complex and interconnected, involving climate factors, vegetation dynamics, and landscape characteristics. Understanding these relationships is crucial for effective water resource management and ecosystem conservation in the face of ongoing land use changes and climate variability (Deng et al. 2015; Su et al. 2023).

6.3 Agricultural practices and irrigation

Various factors influence ET trends in agricultural practices and irrigation. Irrigation methods significantly impact ET rates, with drip irrigation decreasing ET by 13% and transpiration by 11–14% compared to traditional flood irrigation (Wang et al. 2020). This reduction in ET leads to increases in ecosystem and canopy water use efficiency (WUE) of 9–14% and 11%, respectively. Interestingly, the partitioning of ET between evaporation (E) and transpiration (T) can vary due to management activities and environmental conditions. In some cases, E and T can have very similar isotopic signatures, particularly in extreme heat and aridity (Lu et al. 2017). Contrary to expectations, isotopic signatures of E, T, and ET may not always become increasingly enriched as soils dry out, possibly due to root water transport from deeper to shallower soil layers (Lu et al. 2017). In

conclusion, ET trends are influenced by irrigation methods, crop types, and environmental factors. Improved irrigation efficiency, such as drip irrigation, can reduce overall water consumption while increasing crop WUE (Wang et al. 2020). However, the relationship between irrigation efficiency and water conservation is complex, as demonstrated by the paradox of irrigation efficiency (Wellington et al. 2023). Understanding these underlying causes is crucial for developing effective and sustainable water management strategies in agriculture, particularly in regions with limited water resources.

6.4 Urbanization and industrialization

Urbanization and industrialization have a significant impact on ET trends, with urban ET, soil evaporation (E), and vegetation transpiration (T) exhibiting declining trends as urbanization progresses (Chen et al. 2022). This decline can be primarily attributed to changes in urban land use, while increases in urban water body evaporation (W), sensible heat flux (H), and the Bowen ratio (BR) are mainly due to changes in urban microclimate (Chen et al. 2022). Interestingly, the effects of urbanization on water quality and ET are spatially variable. In highly developed watersheds, water quality improvement can accompany urbanization due to the implementation of extensive water management practices. In contrast, less-urbanized watersheds are more sensitive and negatively responsive to land-use changes (Zhao et al. 2015). This spatial variability is influenced by local characteristics, including urbanization levels, environmental management, industrial zone expansion, and land use composition (Zhao et al. 2015). In conclusion, the underlying causes of ET trends during urbanization and industrialization are complex and multifaceted. They involve changes in land use, alterations to the microclimate, and varying levels of environmental management practices. The proposed four-source energy balance model for urban areas (FSU model) facilitates the assessment of the urban heat island (UHI) effect and supports scientific water resource management in urban areas (Li et al. 2017a; Chen et al. 2022). Understanding these trends is crucial for sustainable urban planning and mitigating the negative impacts of rapid urbanization on water resources and the environment.

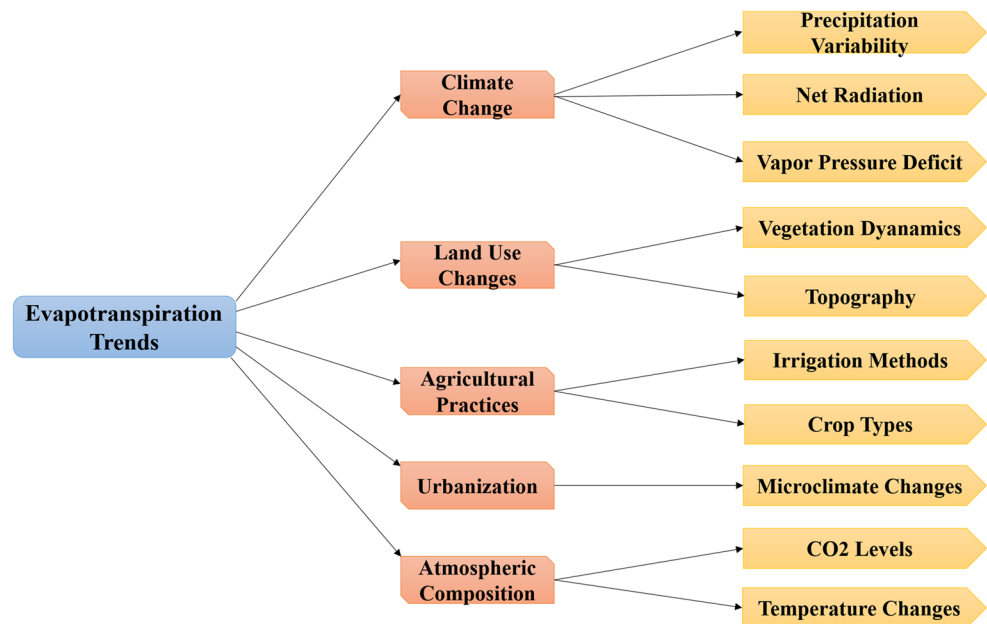
6.5 Changes in atmospheric composition

Changes in atmospheric composition, particularly increasing CO₂ levels, have been identified as a significant driver of global ET trends. The rising CO₂ concentration plays a dominant role in driving variations in apparent underlying water use efficiency (WUE) over time, explaining 66 ± 32% of the total variation (Zhou et al. 2017). This increase in

CO₂ has both direct and indirect effects on ET. The direct effect of rising CO₂ is an increase in potential water-use efficiency (WUE), which parallels the increase in atmospheric CO₂ concentration. However, this effect is partially offset by changes in ecosystem structure, such as increases in leaf area index for different vegetation types (Zhou et al. 2017). Interestingly, the decrease in transpiration due to stomatal closure with rising CO₂ is largely compensated (84%) by an increase in leaf area index, resulting in small changes in the transpiration ratio (Zhou et al. 2017). Contrary to the ET paradox observed in many regions, some tropical areas, such as peninsular Malaysia, have shown an increase in annual and seasonal RET. This increase is primarily attributed to rising minimum temperatures, which were found to be the most influential factor (31.5–48.2%) in defining RET in the region (Pour et al. 2020). The faster rise in minimum temperature after 1981–1985 led to an increase in RET after 1993–1996 in most locations, suggesting a potential link to global warming trends (Pour et al. 2020). In conclusion, while increasing atmospheric CO₂ is a major driver of ET trends globally, its effects are complex and multifaceted. They can be moderated by other factors such as changes in vegetation structure and regional temperature variations. The interplay between these factors highlights the need for comprehensive, region-specific analyses to fully understand ET trends and their implications for water resources and ecosystem functioning in a changing climate.

Figure 6 illustrates the key factors influencing ET trends across India's climatic regions. One of the primary influences is climate variability, particularly in terms of temperature and precipitation patterns. Rising temperatures, especially maximum daily temperatures, have led to increased evaporative demand, notably in arid and semi-arid zones (Adnan et al. 2017a; Vinnarasi et al. 2017). Simultaneously, shifts in rainfall patterns, including delayed monsoons and increased rainfall intensity, alter soil moisture availability, affecting both evaporation and transpiration components. In humid regions, abundant rainfall sustains high ET levels, whereas in semi-arid zones, irregular precipitation limits soil water availability, creating variability in AET (Mo et al. 2018). Solar radiation and wind speed also modulate ET by influencing surface heat balance and vapor transport capacity. However, a declining trend in pan evaporation in some regions has been attributed to reductions in wind speed and changes in atmospheric humidity (Chattopadhyay and Hulme 1997).

In addition to climatic drivers, vegetation dynamics and land-use change play critical roles in shaping ET trends. The Leaf Area Index (LAI), which reflects canopy density, has a strong correlation with ET, especially in cropland and forested regions (Zhang et al., 2014a, b, c). Areas experiencing afforestation or agricultural intensification often

Fig. 6 Influences on Evapotranspiration Trends

see increased ET due to greater transpiration. Conversely, urbanization and deforestation reduce green cover, thus decreasing transpiration and altering the surface energy balance. Irrigation practices also affect ET, as artificial water application increases soil moisture and transpiration, particularly in intensively cultivated regions of northwestern and peninsular India (Srinivas et al. 2013). Furthermore, climate change-induced droughts, as measured by indices like the SPEI, exacerbate ET variability, with deficits in rainfall reducing ET during dry periods (Rehana and Monish 2020). Overall, the interplay between natural climatic processes and anthropogenic land management determines the spatio-temporal variability of ET across India.

7 Implications of ET trends

7.1 Water resource management

ET trends have significant implications for water resource management in the context of climate change. Climate variability affects ET through changes in temperature, precipitation, and other meteorological factors, leading to complex impacts on water availability and hydrological processes (Dimitriadou and Nikolakopoulos 2021; Teklay et al. 2022). In many regions, ET is increasing due to rising temperatures and changes in precipitation patterns. This trend can lead to decreased water availability, particularly in arid and semi-arid areas. For instance, in Egypt, changes in ET have affected agricultural lands and water resources, necessitating adjustments in irrigation practices and water management strategies (Yassen et al. 2020). Similarly, in China,

ET trends have been influenced by both climate change and land-use changes, with significant impacts on water resources and ecosystem services (G. Li et al., 2017a, b). However, the impacts of ET trends on water resources are not uniform across regions. In some cases, increased ET may lead to higher precipitation in certain areas, potentially benefiting water resources. For example, in tropical forests, increased ET can play a beneficial role in local water cycles (Dimitriadou and Nikolakopoulos 2021). It highlights the importance of considering local and regional contexts when assessing the implications of ET trends for water resource management. To address the challenges posed by changing ET patterns, water resource managers need to adopt integrated approaches. These may include implementing nature-based solutions, improving water use efficiency, and developing adaptive management strategies (Nam et al. 2015a; Ciampittiello et al. 2024). Additionally, the use of advanced modelling techniques and climate projections can help in better understanding and predicting future ET trends and their impacts on water resources (Teklay et al. 2022). In conclusion, the implications of ET trends for water resource management are complex and varied. They require careful consideration of local conditions, climate projections, and potential adaptation strategies to ensure sustainable water management in the face of climate change.

7.2 Agricultural productivity

Climate change is projected to have significant impacts on ET trends, with consequent implications for agricultural productivity. ET trends are expected to vary regionally. In South China, agricultural water demand and cropland soil

moisture deficit are projected to decrease, while in North China, water demand and soil moisture deficit are expected to increase (Tao et al. 2003). In South Korea, statistically significant changes in RET have occurred in agricultural areas since the 1970s, particularly during winter months (Nam et al. 2015b). Egypt has experienced significant changes in the spatial distribution of RET over the last 35 years, with the northeastern regions and older agricultural lands being the most affected (Yassen et al. 2020). These ET trends have important implications for crop water requirements and agricultural productivity. In Mediterranean countries, maize, wheat, and grape production are projected to require 13%, 16%, and 10% more water, respectively, under future climate scenarios (Masia et al. 2021). This increased water demand poses challenges for irrigation and water resource management. However, conservation agriculture practices, such as no-till with stubble retention, can help improve soil water storage and water productivity for crops like maize, wheat, and soybeans (Yang et al. 2018). To adapt to changing ET patterns, various water management strategies show promise. Deficit irrigation, when applied strategically, can improve water productivity while limiting yield losses in maize (Allakonon et al. 2022). Conservation agriculture, combined with drip irrigation, has been found to reduce ET and irrigation water use in Sub-Saharan Africa significantly (Assefa et al. 2018). Climate-smart agriculture approaches have also shown potential to mitigate the impacts of climate change and boost productivity (Okolie et al. 2022). Overall, adapting irrigation and farming practices to regional trends in ET will be crucial for maintaining agricultural productivity in a changing climate.

7.3 Ecosystem health and biodiversity

Based on the provided context, there is no direct information about ET trends and their implications for ecosystem health and biodiversity. However, I can provide some relevant insights from the given papers that relate to ecosystem health, biodiversity, and environmental changes: Ecosystem health and biodiversity are closely interconnected, with biodiversity playing a crucial role in maintaining ecosystem stability and resilience. Studies have shown that higher species richness and functional diversity contribute to increased temporal stability in ecosystem productivity (Morin et al. 2014). This stability is primarily driven by compensatory dynamics across species, supporting the biodiversity insurance hypothesis (Morin et al. 2014). The asynchrony of species responses to small disturbances, rather than environmental fluctuations, is the main factor behind this pattern (Morin et al. 2014). Interestingly, research has shown that changes in biodiversity, caused by various environmental drivers, have a consistent impact on ecosystem stability. A

given decrease in plant species numbers leads to a quantitatively similar decrease in ecosystem stability, regardless of the specific driver causing the biodiversity loss (Hautier et al. 2015). It highlights the importance of maintaining biodiversity for ecosystem resilience. While the papers do not specifically address ET trends, they emphasize the need for a holistic approach to ecosystem management. Maintaining sufficient levels of biodiversity is crucial for ensuring ecosystem resilience against multiple and stochastic disturbances (Sasaki et al. 2015). This resilience-based management approach focuses on conserving key elements of biodiversity that confer ecosystem resilience, such as functional redundancy, response diversity, and keystone species (Walker 1995; Sasaki et al. 2015). In conclusion, although the direct implications of ET trends are not addressed in the provided context, the research underscores the importance of preserving biodiversity and ecosystem functions in the face of environmental changes. Future studies should investigate the specific impacts of ET trends on ecosystem health and biodiversity, informing more targeted conservation strategies.

7.4 Climate modeling and weather prediction

ET trends have significant implications for climate modeling and weather prediction. Studies have shown that ET has increased over time, partly due to vegetation growth, which acts as a cooling mechanism and may slow the climate-warming trend (Kramer et al. 2015). This vegetation negative feedback varies between models, highlighting the complexity of incorporating ET dynamics into climate projections. Climate change scenarios consistently predict increases in PET rates, which would significantly affect catchment water balances (Gharbia et al. 2018). However, the sensitivity of ET to climate variables differs across regions and seasons. For instance, in Mediterranean climates, ET is more sensitive to changes in spring and summer precipitation than to vapor pressure deficit scenarios (Montaldo and Oren 2018). This variability underscores the importance of considering regional and seasonal factors in climate models. In conclusion, an accurate representation of ET trends is crucial for improving climate models and weather forecasts. The use of advanced techniques, such as GIS-based algorithms for fine-scale, spatially distributed PET modelling (Gharbia et al. 2018) and causal inference models for soil moisture-precipitation coupling (Li et al. 2020), can enhance our understanding of land-atmosphere interactions. These advancements are crucial for improving the accuracy of climate projections, attributing past climate changes, and assessing climate models (Deser and Phillips 2023).

Figure 7 outlines the broad implications of ET trends across India's diverse climatic zones, highlighting their

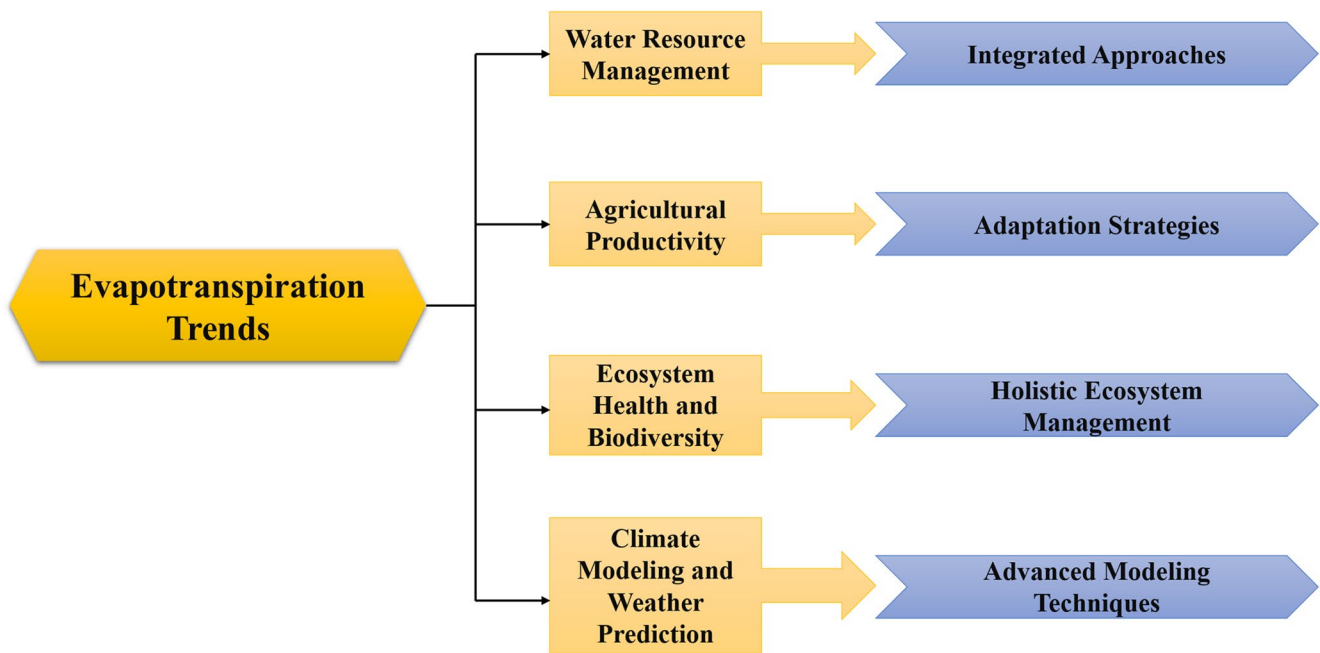


Fig. 7 Implications of Evapotranspiration Trends

critical influence on agriculture, water resource management, and ecosystem sustainability. One major implication lies in agricultural productivity, as ET directly affects soil moisture availability and plant water uptake. Regions experiencing declining ET, especially due to reduced rainfall or vegetation cover, face increased risks of crop stress and yield losses (Rehana and Monish 2020). On the other hand, rising ET in water-scarce zones can aggravate irrigation demands, placing further pressure on already limited freshwater resources (Mo et al. 2018). These changes necessitate region-specific irrigation planning, crop diversification, and the adoption of water-efficient farming practices. ET variability also influences drought characterization; for instance, high ET coupled with low rainfall intensifies meteorological and agricultural droughts, as captured by indices like SPEI and SPI (Adnan et al. 2017a).

Beyond agriculture, ET trends carry important implications for climate modeling, water budgeting, and forest management. In hydrological modeling, underestimating ET can lead to erroneous runoff estimates and flawed water balance calculations (Srinivas et al. 2013). Regions with rapidly increasing ET may witness declining groundwater recharge and shrinking surface water bodies, threatening long-term water security (Vinnarasi et al. 2017). From an ecological perspective, ET trends impact carbon cycling and land-atmosphere interactions, as vegetation-mediated transpiration influences atmospheric humidity and local rainfall feedbacks (Zhang et al., 2014a, b, c). In forested regions, shifts in ET may also signal ecosystem stress or resilience, depending on vegetation response to climate drivers.

Policymakers and resource managers must therefore integrate ET trends into climate adaptation frameworks, land-use planning, and drought mitigation strategies to ensure sustainable environmental governance across India's vulnerable landscapes.

8 Knowledge gaps and future research directions

8.1 Limitations of current studies

Current studies on ET trends across different climatic zones in India have several limitations: The use of PET instead of AET in drought assessments may not accurately capture regional variations, especially in water-limited areas. AET accounts for both water and energy-based evaporative demands, providing a more comprehensive understanding of drought characteristics in different climatic zones (Rehana and Monish 2020). Many studies rely on the SPEI for drought assessment, which may not adequately capture the strong seasonality of rainfall in India. Restructuring SPEI to account for water availability deficit rather than atmospheric water demand could improve drought characterization in various meteorological homogeneous zones of India (Rehana and Monish 2020). There is a lack of consistency in results across different drought indices and PET calculation methods. For example, in China, precipitation and SPEI indicated wetting trends at the national scale, while all Palmer Drought Severity Index (PDSI)

variants showed drying trends (Chen et al. 2017). This inconsistency underscores the need for a more comprehensive approach to evaluating ET trends across various climatic zones in India. Regional factors such as aerosol concentration, greenhouse gases, and land cover changes are not always adequately considered in ET trend analyses. These factors can significantly affect the space-time variability of rainfall at the local scale, which in turn influences ET trends (Pal et al. 2019). The interplay between biogeochemical and biogeophysical processes in land-use changes is often overlooked. Similar land-use changes can have varied, even reverse, biogeophysical forcing on local temperatures across different climate regimes, affecting ET trends (Huang et al. 2018). In conclusion, future studies on ET trends across different climatic zones in India should consider using AET instead of PET, restructure drought indices to better capture seasonal variability, employ multiple indices for robust results, account for regional factors that affect rainfall variability, and consider the complex interactions between land-use changes and local climate. These improvements would provide a more comprehensive understanding of regional variations in ET trends and their underlying causes.

8.2 Emerging technologies for ET measurement

Remote sensing has emerged as a cost-effective method for estimating ET at regional to global scales, overcoming the inherent difficulties in directly measuring ET, especially over large areas (Zhang et al. 2016a). This technology enables robust ET estimates through the application of data assimilation and fusion techniques, as well as the utilization of multi-source remote sensing data and the latest sensor technologies (Zhang et al. 2016a). Interestingly, while remote sensing technologies are advancing, studies have shown contradictory trends in ET across different regions and time scales. In India, both pan evaporation and PET have decreased in recent years (Chattopadhyay and Hulme 1997). However, future warming scenarios suggest a likely increase in PET over India, although this increase is expected to be unequal between regions and seasons (Chattopadhyay and Hulme 1997). The standardized precipitation actual evapotranspiration index (SPAETI), which uses AET estimated from the Budyko framework and remote sensing data, has been proposed as a more robust drought assessment tool for water-limited regions in India (Rehana and Monish 2020). In conclusion, emerging technologies like remote sensing are improving our ability to measure and estimate ET across large scales. However, the trends in ET across different climatic zones in India are complex and sometimes contradictory. While some studies show decreasing trends in pan evaporation and PET, future projections

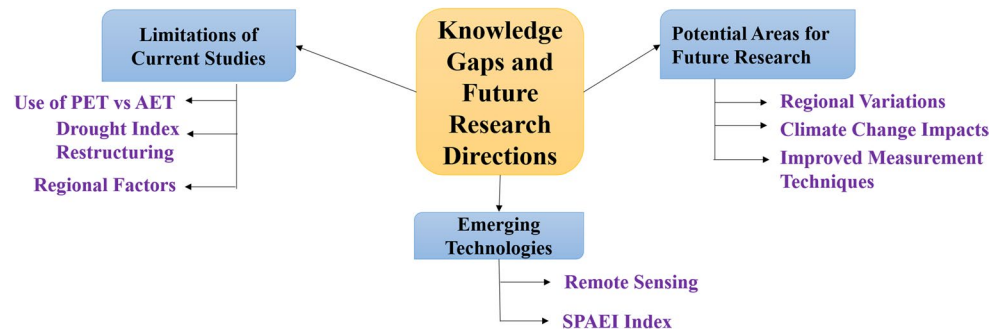
suggest potential increases in these values. The use of indices like SPAETI, which incorporate AET, may provide more accurate assessments of drought conditions in water-limited regions of India.

8.3 Potential areas for future research

Future research on ET measurement and trends across different climatic zones in India could focus on several key areas. Regional variations in ET trends require further investigation, as studies have shown contradictory results across various parts of India. For instance, while pan evaporation and PET have decreased in recent years at the national scale (Chattopadhyay and Hulme 1997), there are likely significant regional differences that warrant closer examination. Research could explore how ET trends vary across India's diverse climatic zones, from arid to humid regions (Chen et al. 2017). A key area for future work is the enhancement of ET measurement and modelling techniques to account for India's complex topography and climate. Studies could compare different methods for calculating PET, such as the Penman-Monteith and Thornthwaite equations, to determine which are most suitable for different regions of India (Xu et al. 2014; Chen et al. 2017). Additionally, research could investigate how to incorporate interactions between climatic variables in ET models, as these are significant in other regions (Xu et al. 2014). Future studies should examine the impacts of climate change on ET patterns across India. It could include projecting future scenarios of PET under different climate change models, as has been done for other countries (Chattopadhyay and Hulme 1997; Xu et al. 2014). Such research is crucial for understanding the potential impacts on water resources, agriculture, and ecosystems in various parts of India. In summary, there is a need for more comprehensive, region-specific studies on ET trends and measurement techniques across India's varied climatic zones. This research would provide valuable insights for water resource management and climate change adaptation strategies in the country.

Figure 8 highlights the critical knowledge gaps and future research directions in ET studies, especially within the Indian context. A major limitation in current ET research is the scarcity of high-resolution and long-term ground-based observations, which hampers accurate trend analysis and model validation (Dimitriadou and Nikolakopoulos 2021). While satellite-based ET products have improved, they still require extensive calibration and validation, particularly in heterogeneous landscapes such as India's agro-climatic zones (Mo et al. 2018). Additionally, there is a need to disaggregate ET components—such as soil evaporation, plant transpiration, and canopy interception—to better understand the interactions between vegetation and climate. Despite the

Fig. 8 Knowledge Gaps and Future Research Directions in ET Studies



known impacts of climate drivers, such as temperature and radiation, as well as land-use change, urbanization, and CO₂ fertilization effects on ET, these remain underexplored in Indian studies (Lemordant et al. 2018). These knowledge gaps highlight the necessity for integrated studies that combine satellite data, in situ measurements, and climate modelling.

Moving forward, future research should prioritize regional-scale modelling of ET under different climate scenarios, particularly using dynamic vegetation models and downscaled climate projections (Wanniarachchi and Sarukkalgige 2022). There is also a pressing need to improve drought prediction models by incorporating ET variability in conjunction with indices like SPI and SPEI (Rehana and Monish 2020). Furthermore, interdisciplinary approaches combining hydrology, climatology, remote sensing, and socio-economic analysis will be crucial to assess how ET trends impact water security, agricultural resilience, and ecosystem services. Incorporating machine learning algorithms for spatio-temporal ET pattern recognition and trend attribution is another promising direction (Mannan et al. 2018). To support adaptive policy and decision-making, future research should also focus on developing region-specific ET estimation tools and guidelines for water resource planning, especially in data-scarce regions. These advancements will collectively strengthen India's preparedness for climate variability and long-term water management.

9 Discussions and summary

ET is a critical component of the hydrological cycle, integrating water evaporation from land and water bodies with plant transpiration. It plays a pivotal role in climate regulation, agriculture, and water resource management. Understanding ET trends is crucial for assessing the impacts of climate change, enhancing drought prediction, and ensuring sustainable water availability. Despite a general increase in temperatures due to global warming, some regions report a decrease in ET. This counterintuitive trend can be attributed to land-use changes and phenomena like solar dimming,

while monsoonal variability remains a dominant force influencing ET dynamics across the Indian subcontinent.

9.1 Methodology for ET measurement

ET is measured using various methods. The Penman-Monteith equation is widely regarded as the most accurate, especially for calculating RET. Remote sensing techniques enable large-scale monitoring, whereas traditional methods, such as pan evaporation measurements, are less reliable. More precise but localized approaches include lysimeters and eddy covariance systems. For analysing ET trends, statistical techniques such as the Mann-Kendall and Spearman's Rho tests are commonly used. Additionally, machine learning models, such as Random Forest, have shown promise in capturing spatial and temporal variations in ET.

9.2 ET trends across india's climatic zones

India's diverse climatic zones exhibit significant variation in ET behaviour. In the Tropical Wet Zone, ET is influenced by monsoon seasonality and is often reduced due to persistent cloud cover. The Tropical Dry Zone experiences elevated ET rates during prolonged dry seasons, largely due to high vapor pressure deficits. Subtropical Humid Zones display mixed ET trends, with some regions witnessing declines owing to reduced solar radiation. In the Montane Zones, especially the Himalayas, ET typically decreases at higher altitudes, influenced by reduced temperatures and radiation. Conversely, Arid and Semi-Arid Zones report increasing ET trends, driven by rising temperatures and altered land use.

9.3 Regional variations in ET trends

ET patterns also vary regionally within India. North India, particularly the Gangetic Plains, shows increased ET, closely tied to seasonal fluctuations. South India presents a contrasting picture, with both rising and declining ET trends attributed to heterogeneous temperature changes. East India, heavily influenced by the monsoon, is increasingly experiencing droughts, which affect ET. In West India, improved water

management and a decline in evaporative demand have led to a reduction in drought frequency. CI shows mixed results, with ET either increasing or decreasing, depending on local changes in solar radiation and wind patterns.

9.4 Underlying causes of ET trends

Several factors influence these ET variations. Climate change, particularly global warming, has region-specific effects on ET. Land-use changes, such as deforestation and urbanization, significantly alter ET dynamics. Modern agricultural practices, including the adoption of drip irrigation, help reduce ET compared to traditional flood irrigation. Urbanization and industrialization lead to declines in ET due to impervious surfaces and vegetation loss. Additionally, changes in atmospheric composition, particularly rising CO₂ concentrations, impact plant physiology and transpiration rates, further influencing ET (Lemordant et al. 2018).

9.5 Implications of ET trends

Changes in ET have far-reaching implications. Water resource management strategies must be adapted to accommodate ET-driven shifts in groundwater recharge and surface water availability. In agriculture, understanding ET fluctuations is crucial for aligning crop water requirements with irrigation strategies. Ecosystems are also affected, as ET influences local microclimates and plant growth. Accurate representation of ET in climate and weather models is crucial for enhanced forecasting, thereby enabling policymakers and stakeholders to make informed decisions.

10 Future research directions

Future research should focus on enhancing ET measurement by integrating remote sensing data and AI-based models to improve accuracy and precision. A deeper understanding of the interactions between ET and extreme weather events is essential. Developing region-specific water management and climate adaptation strategies will require spatially explicit ET datasets and improved model simulations. There is also a need for interdisciplinary approaches combining climatology, hydrology, remote sensing, and socio-economic dimensions to fully understand the implications of changing ET patterns in India.

10.1 Importance of continued monitoring and analysis

Continued monitoring and analysis of ET across India's diverse climatic regions is crucial for sustainable resource

management. Crop RET serves as a foundational variable for estimating crop water requirements (Poddar et al. 2021). In the sub-humid subtropical Western Himalayan region, temperature-based ET₀ methods outperform radiation- and pan evaporation-based methods, underlining the importance of localized calibration. Solar radiation, maximum temperature, and relative humidity emerge as key inputs for accurately estimating ET₀ in this region, indicating that regional climatic characteristics should guide the selection of input parameters for ET models (Du et al. 2024). Understanding spatial and temporal variations in ET helps optimize agricultural practices and manage water resources effectively. As climate change progresses, changes in ET will significantly impact irrigation strategies and crop yields. Research must therefore continue to refine ET estimation techniques and build robust, regionally adaptable models. Long-term data monitoring can support the formulation of proactive water policies, enabling better adaptation to shifting weather patterns (Nam et al. 2015b).

10.2 Recommendations for policymakers and researchers

To address India's complex ET dynamics, several recommendations can be made. First, ET measurement methods should be chosen carefully, considering both accuracy and data availability. While the FAO-56 Penman-Monteith method is widely accepted for estimating ET₀ across India's climatic zones (Nandagiri and Kovoov 2006), traditional methods continue to be used due to their simplicity and ease of application. Policymakers should invest in data collection infrastructure to promote the adoption of the more accurate FAO method. Interestingly, declining trends in pan evaporation and PET observed in India challenge the assumption that climate warming will always lead to increased ET (Chattopadhyay and Hulme 1997). Researchers must investigate these anomalies to ensure sound water resource planning.

For drought assessment, indices such as the SPEI, which incorporates AET data, are particularly useful in arid and semi-arid regions (Rehana and Monish 2020). These indices account for both energy- and water-based constraints, making them more reliable in regions with high rainfall seasonality. Policymakers should consider integrating such comprehensive indices into national drought monitoring systems. In conclusion, strengthening ET measurement techniques, understanding regional disparities, and adopting advanced indices, such as SPAEI, will collectively enhance India's climate resilience and water resource governance.

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Declarations

Competing interests The authors declare no competing interests.

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